$$= \frac{(p+q)^2 \pm \sqrt{(p+q)^2\left[(p+q)^2 - 4pq\right]}}{2pq}$$

$$= \frac{(p+q)^2 \pm \sqrt{(p+q)^2(p^2+q^2+2pq-4pq)}}{2pq}$$

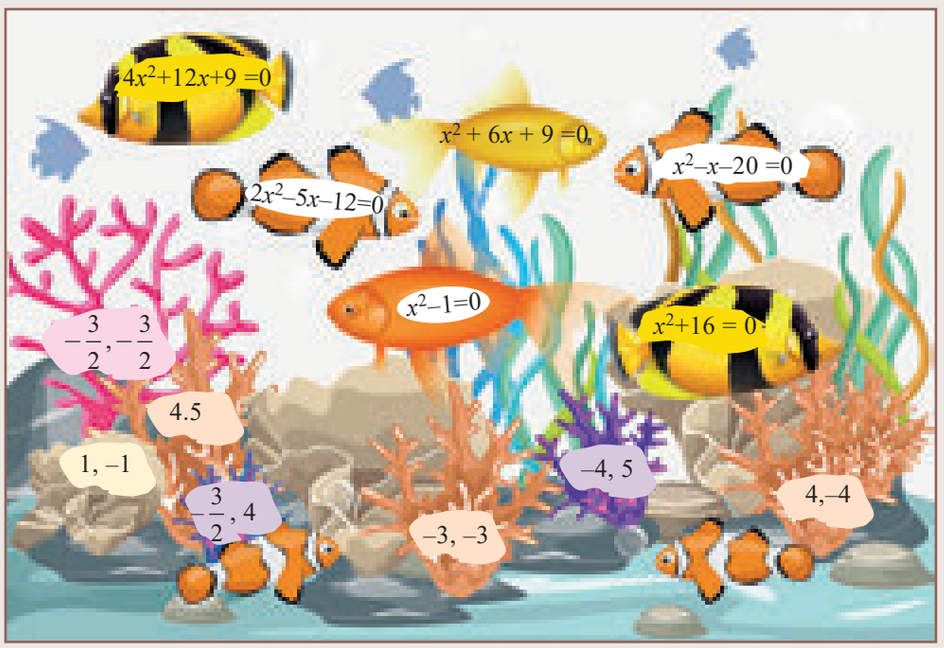
$$= \frac{(p+q)^2 \pm \sqrt{(p+q)^2(p-q)^2}}{2pq}$$

$$= \frac{(p+q)^2 \pm (p+q)(p-q)}{2pq} = \frac{(p+q)\{(p+q) \pm (p-q)\}}{2pq}$$

Therefore, $x = \frac{p+q}{2pq} \times 2p$, $\frac{p+q}{2pq} \times 2q$ we get, $x = \frac{p+q}{q}$, $\frac{p+q}{p}$

Activity 3

Serve the fishes (Equations) with its appropriate food (roots). Identify a fish which cannot be served?



Exercise 3.11

1. Solve the following quadratic equations by completing the square method

   (i) $9x^2 - 12x + 4 = 0$ (ii) $\frac{5x+7}{x-1} = 3x + 2$

2. Solve the following quadratic equations by formula method

   (i) $2x^2 - 5x + 2 = 0$ (ii) $\sqrt{2}f^2 - 6f + 3\sqrt{2} = 0$ (iii) $3y^2 - 20y - 23 = 0$

   (iv) $36y^2 - 12ay + (a^2 - b^2) = 0$

3. A ball rolls down a slope and travels a distance $d = t^2 - 0.75t$ feet in t seconds. Find the time when the distance travelled by the ball is 11.25 feet.

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